## benchmark.cpp

```
Starting Benchmarks using omp_get_wtime()...
[DONE] Merge Sort
[DONE] Matrix Multiplication
[DONE] BFS

All benchmarks executed successfully using omp_get_wtime()! Output exported to:
F:\BRACU\CSE706 - Parallel Algorithm\Parallel_Computing_Project\benchmark_results.csv
```

## collapse_benchmark.cpp

```
Running Matrix Multiplication Collapse Benchmark...

Matrix    Threads   Normal parallel for   collapse(2)      Faster
-------------------------------------------------------------------------
256       1         679.999828 ms         848.999977 ms    Normal
256       2         411.000013 ms         392.999887 ms    Collapse(2)
256       4         233.999968 ms         226.999998 ms    Collapse(2)
256       8         130.000114 ms         128.999949 ms    Collapse(2)
512       1         4372.999907 ms        6392.999887 ms   Normal
512       2         3378.999949 ms        3419.000149 ms   Normal
512       4         2331.999779 ms        2236.999989 ms   Collapse(2)
512       8         1194.000006 ms        1196.000099 ms   Normal
1024      1         43437.000036 ms       46072.999954 ms  Normal
1024      2         27975.999832 ms       28115.000010 ms  Normal
1024      4         18015.999794 ms       17832.999945 ms  Collapse(2)
1024      8         9400.000095 ms        9437.999964 ms   Normal

[SUCCESS] Benchmark results saved to CSV: data\benchmark_results\collapse_results.csv
```

## controlled_matrix_benchmark.cpp

```
=========================================================================
 CONTROLLED MATRIX MULTIPLICATION BENCHMARK (Identical Input Data)
=========================================================================

>>> Generating Controlled Master Input Matrices A and B (256x256)... Done.

Matrix    Threads   Sequential      Static          Dynamic         Collapse(2)
-----------------------------------------------------------------------------------
256       1         809.000015 ms   661.999941 ms   784.000158 ms   597.999811 ms
256       2         809.000015 ms   465.999842 ms   460.000038 ms   465.000153 ms
256       3         809.000015 ms   319.000006 ms   307.999849 ms   319.000006 ms
256       4         809.000015 ms   251.000166 ms   282.999992 ms   221.999884 ms

>>> Generating Controlled Master Input Matrices A and B (512x512)... Done.

Matrix    Threads   Sequential      Static          Dynamic         Collapse(2)
-----------------------------------------------------------------------------------
```

```
512     1       6111.999989 ms  9383.999825 ms  7073.000193 ms  6431.999922 ms
512     2       6111.999989 ms  3729.000092 ms  3820.999861 ms  3741.999865 ms
512     3       6111.999989 ms  2687.000036 ms  2556.999922 ms  2653.000116 ms
512     4       6111.999989 ms  2209.000111 ms  2206.999779 ms  2128.000021 ms

[SUCCESS] Controlled benchmark results logged to: data\benchmark_results\controlled_matrix_results.csv
```

**full_benchmark.cpp**

```
================================================================================
 WEEK 3 MAJOR BENCHMARK EXPERIMENT (3 Sizes x 4 Threads = 12 Runs per Variant)
================================================================================

>>> Generating Matrix Data N=256 (256x256)... Done.
>>> Executing Ground Truth Sequential Baseline... 613.00 ms

Matrix  Threads  Sequential    Static      Dynamic     Collapse(2)   Verified
--------------------------------------------------------------------------------
256     1       612 ms       559 ms       591 ms       495 ms       PASSED
256     2       612 ms       460 ms       364 ms       373 ms       PASSED
256     4       612 ms       289 ms       285 ms       284 ms       PASSED
256     8       612 ms       189 ms       159 ms       158 ms       PASSED

>>> Generating Matrix Data N=512 (512x512)... Done.
>>> Executing Ground Truth Sequential Baseline... 5852.00 ms

Matrix  Threads  Sequential    Static      Dynamic     Collapse(2)   Verified
--------------------------------------------------------------------------------
512     1       5851 ms      5461 ms      5723 ms      4891 ms      PASSED
512     2       5851 ms      3369 ms      3251 ms      3463 ms      PASSED
512     4       5851 ms      2344 ms      2298 ms      2275 ms      PASSED
512     8       5851 ms      1279 ms      1328 ms      1372 ms      PASSED

>>> Generating Matrix Data N=1024 (1024x1024)... Done.
>>> Executing Ground Truth Sequential Baseline... 47559.00 ms

Matrix  Threads  Sequential    Static      Dynamic     Collapse(2)   Verified
--------------------------------------------------------------------------------
1024    1       47559 ms     46868 ms     49139 ms     46720 ms     PASSED
1024    2       47559 ms     28425 ms     26924 ms     28741 ms     PASSED
1024    4       47559 ms     18789 ms     17985 ms     18138 ms     PASSED
1024    8       47559 ms     9973 ms      9896 ms      10082 ms     PASSED

================================================================================
[SUCCESS] Week 3 full benchmark complete. All 12 test points passed verification.
Dataset written to: data\benchmark_results\week3_full_benchmark.csv
================================================================================
```

**matrix_schedule_benchmark.cpp**

```
=== Matrix Multiplication Schedule Benchmark (512x512) ===
```

```
--- Thread Count: 1 ---
Threads: 1 | Schedule: static | Time: 5444 ms
Threads: 1 | Schedule: dynamic | Time: 5319 ms
Threads: 1 | Schedule: guided | Time: 5222 ms

--- Thread Count: 2 ---
Threads: 2 | Schedule: static | Time: 3684 ms
Threads: 2 | Schedule: dynamic | Time: 3378 ms
Threads: 2 | Schedule: guided | Time: 3968 ms

--- Thread Count: 3 ---
Threads: 3 | Schedule: static | Time: 2972 ms
Threads: 3 | Schedule: dynamic | Time: 2858 ms
Threads: 3 | Schedule: guided | Time: 2852 ms

--- Thread Count: 4 ---
Threads: 4 | Schedule: static | Time: 2316 ms
Threads: 4 | Schedule: dynamic | Time: 2345 ms
Threads: 4 | Schedule: guided | Time: 2331 ms
```

**test_race_conditions.cpp**

```
Testing Parallel Merge Sort for Race Conditions (10 runs, 4 threads)...

Run 1 -> Correct
Run 2 -> Correct
Run 3 -> Correct
Run 4 -> Correct
Run 5 -> Correct
Run 6 -> Correct
Run 7 -> Correct
Run 8 -> Correct
Run 9 -> Correct
Run 10 -> Correct

----------------------------------------
SUCCESS: All 10 runs sorted correctly. No race conditions detected!
```

**verified_matrix_benchmark.cpp**

```
======================================================================
 STRICT VERIFICATION & BENCHMARKING (Sequential == Parallel Check)
======================================================================

>>> Initializing Master Input Matrices A and B (256x256)... Done.
>>> Running Sequential Baseline... 726 ms

Matrix  Threads Sequential     Static        Dynamic       Collapse(2)    Verified
-------------------------------------------------------------------------------------
256     1       726.000071 ms  638.999939 ms  628.000021 ms  552.000046 ms  PASSED
256     2       726.000071 ms  471.999884 ms  417.999983 ms  452.999830 ms  PASSED
```

[CORRUPTION DETECTED] Collapse(2) output mismatch at N=256, threads=3!

**bfs_parallel.cpp**

BFS (Parallel Level-Synchronous) Time: 1158.64 ms

**matrix_multiplication_parallel.cpp**

Matrix Multiplication (Parallel For) Time: 1635.32 ms

**bfs_sequential.cpp**

Enter the number of vertices: 6
Enter the number of edges: 5

Enter 5 edges (u v pairs, 0-indexed):
0 1
0 2
1 3
1 4
2 5

Enter the starting source node for BFS: 0

BFS Traversal starting from node 0: 0 1 2 3 4 5

**merge_sort_sequential.cpp**

Enter the number of elements: 5
Enter 5 space-separated elements: 25 4 85 22 6

Original array: 25 4 85 22 6
Sorted array: 4 6 22 25 85

**matrix_multiplication_sequential.cpp**

Enter rows and columns for Matrix A (R1 C1): 3 3
Enter rows and columns for Matrix B (R2 C2): 3 3

Enter elements of Matrix A (3x3):
1 2 3
4 5 6
7 8 9

Enter elements of Matrix B (3x3):
11 22 33
44 55 66
77 88 99

Resultant Matrix C (3x3):
330 396 462
726 891 1056

1122 1386 1650